3D Creative Club Study Guide

Model a Snowflake

Goal: What You Will Build

Object Mode



Steps:

- Model one branch of a snowflake using Edit Mode
- Use the Mirror modifier to copy work across an axis
- Use the Array modifier with Object Offset to make a 6-fold pattern
- Apply modifiers and merge duplicate vertices
- Add thickness with Solidify and smooth edges with Bevel
- After the tutorial, use your imagination to design your own snowflake!

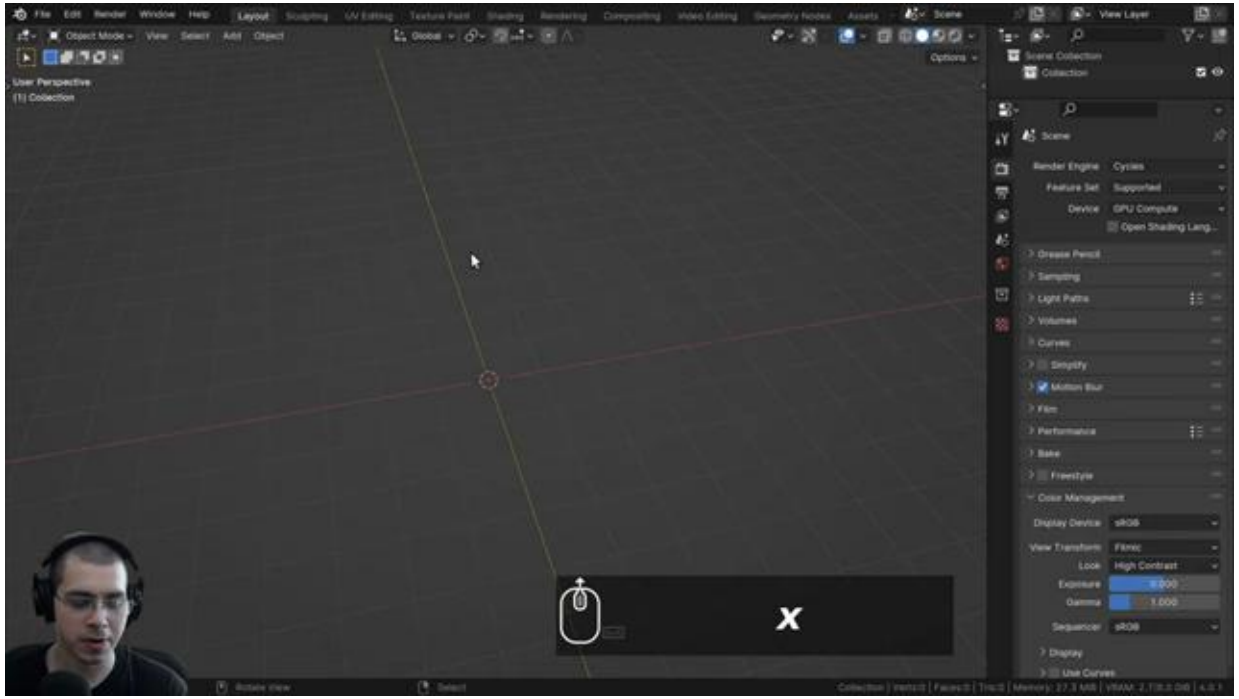
Shortcuts:

Shift + A: Add · Tab: Mode toggle · G/S/R: Transform

Tip: This week you learn modifiers — non-destructive ways to repeat and mirror geometry.

Step 1: Start with a Plane

Object Mode



Steps:

- Press X to delete the default cube
- Shift + A → Mesh → Plane
- Press Tab to enter Edit Mode
- Use the front view (`) to work straight-on

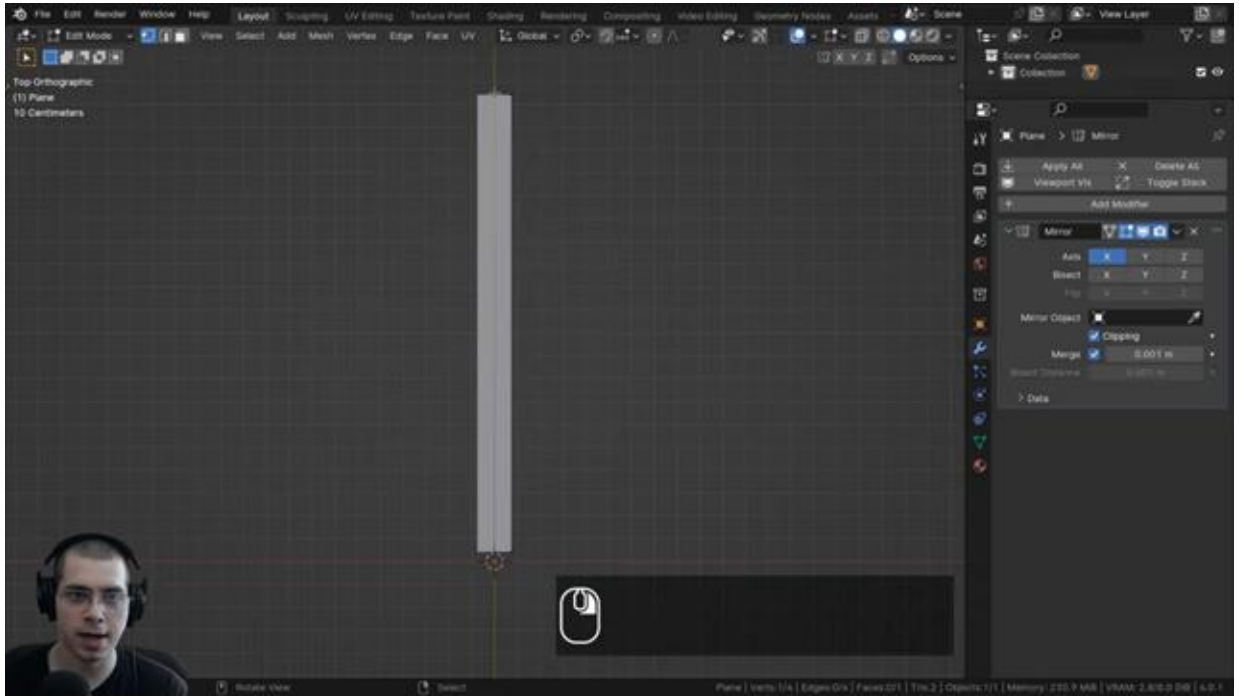
Shortcuts:

X: Delete · Shift + A: Add · Tab: Edit Mode · `: View selector

Tip: A flat plane is the perfect starting point for one snowflake branch.

Step 2: Model the Main Branch

Edit Mode



Steps:

- Scale the plane into a long, thin rectangle (S, Y)
- Subdivide the plane to create more vertices (Right-click → Subdivide)
- Add a Mirror modifier across the Y axis
- This means editing one side mirrors changes to the other side

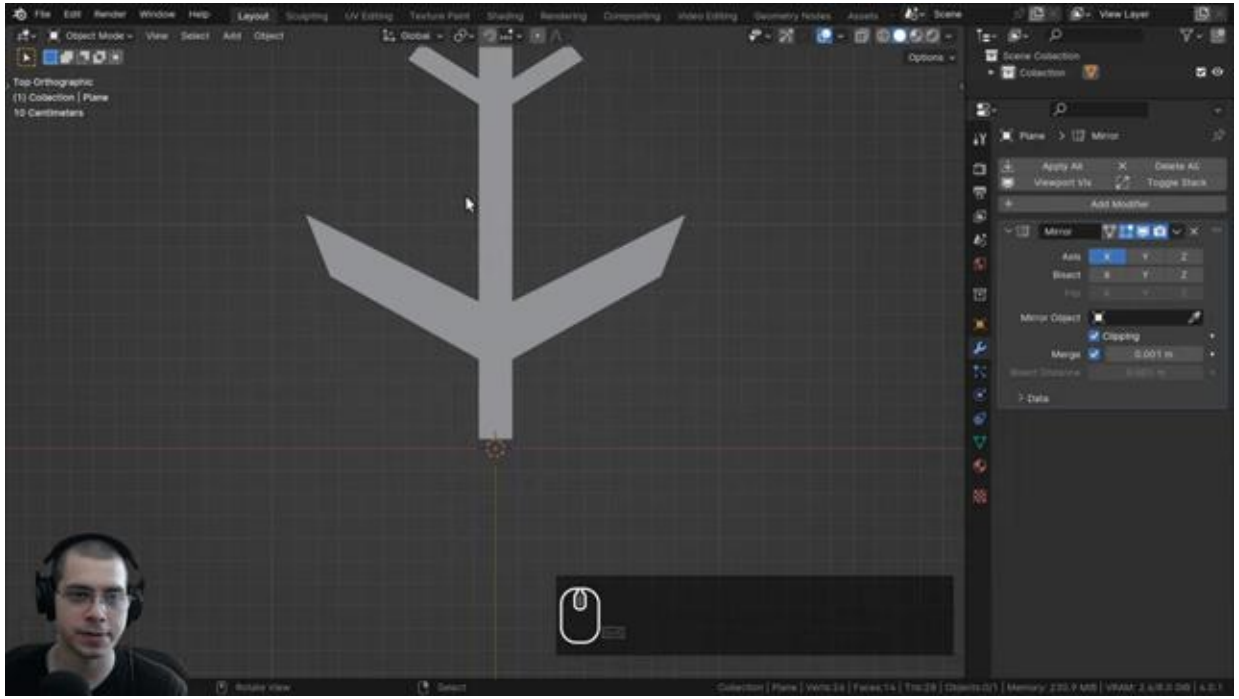
Shortcuts:

S: Scale · Ctrl + R: Loop cut · Tab: Object Mode to add modifier

Tip: Mirror is the secret to symmetrical snowflakes: model 1/2, mirror does the rest.

Step 3: Add Side Branches

Edit Mode



Steps:

- Select vertices along the main branch
- Extrude out side branches (E, then move)
- Scale the branch tips smaller for a natural look
- Add smaller V-shaped notches near the tips

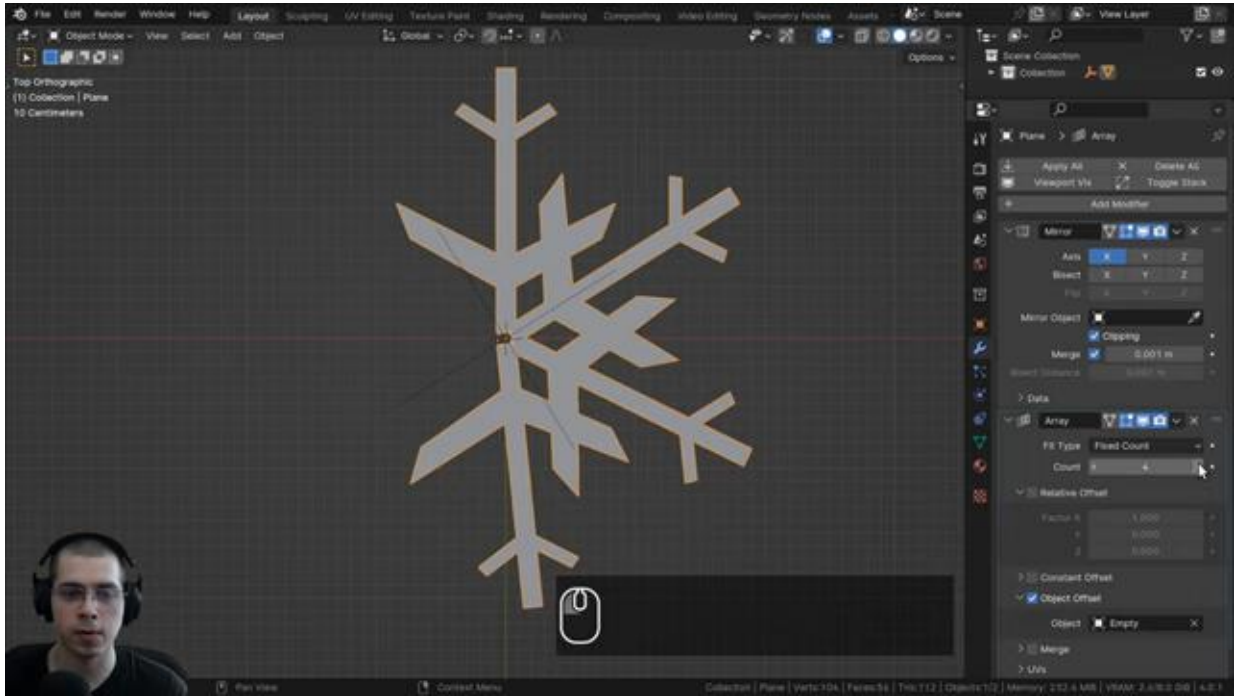
Shortcuts:

E: Extrude · S: Scale · G: Move · Ctrl + Tab: Mesh select mode

Tip: Real snowflakes have repeating shapes at different sizes. Keep it simple but detailed.

Step 4: Create the Circular Array

Modifiers



Steps:

- Return to Object Mode
- Add an Empty at the center (Shift + A → Empty → Plain Axes)
- Add an Array modifier to the branch
- Set Count to 6 and enable Object Offset
- Pick the Empty as the Object Offset target
- Rotate the Empty 60° on Z to create 6 copies

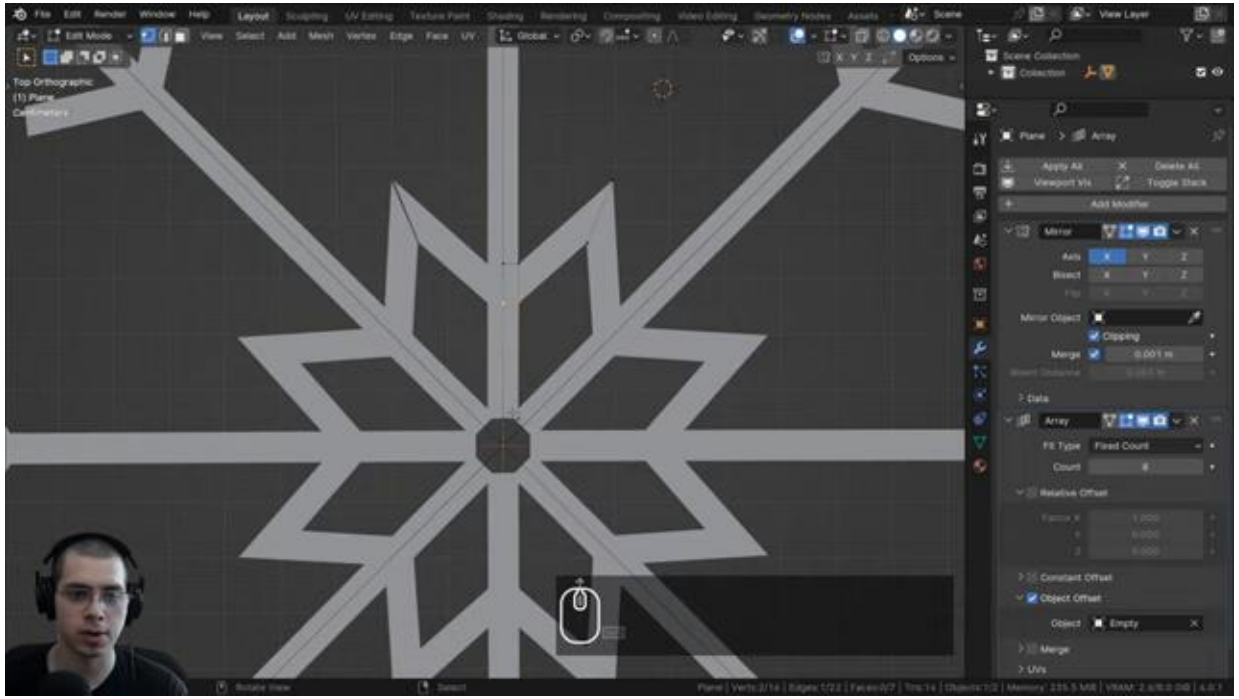
Shortcuts:

Shift + A: Empty · R + Z + 60: Rotate 60° · `: Top view

Tip: 6 copies at 60° each = 360°. This is why snowflakes have 6-fold symmetry.

Step 5: Refine the Pattern

Modifiers



Steps:

- Look at the full 6-branch snowflake
- Tab back into Edit Mode to tweak branch shapes
- Add more cuts or notches to make the pattern interesting
- The Array modifier updates all 6 copies live

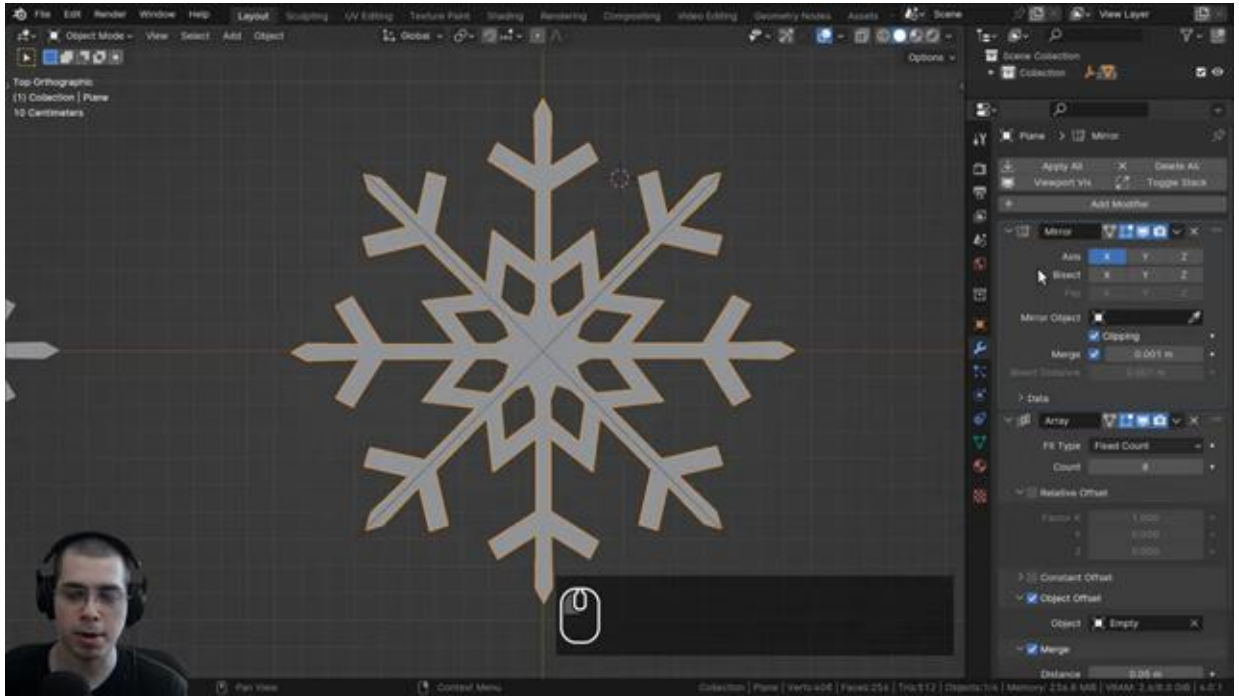
Shortcuts:

Tab: Edit Mode · Ctrl + R: Loop cut · E: Extrude

Tip: Because the Array is still active, any change to one branch updates the whole snowflake.

Step 6: Apply Modifiers & Merge

Object Mode



Steps:

- When you are happy with the shape, apply the Array modifier
- Apply the Mirror modifier too
- Tab into Edit Mode
- Select all and press M → By Distance to merge overlapping vertices

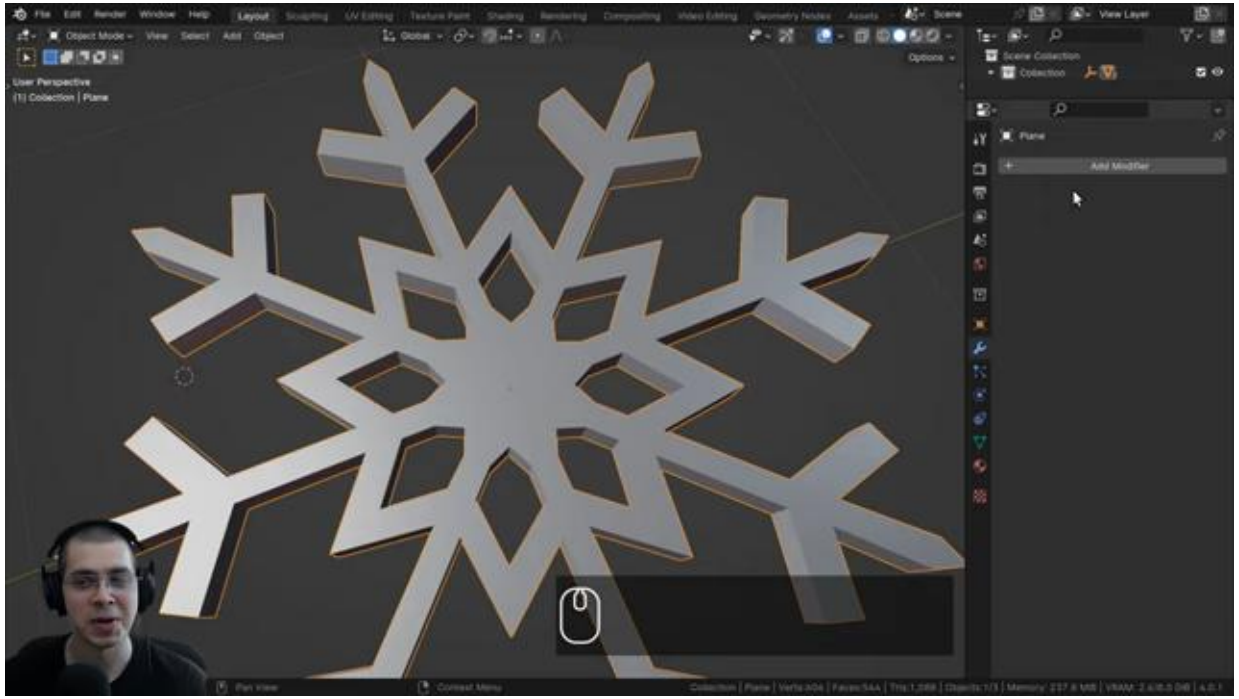
Shortcuts:

Apply: click down arrow on modifier → Apply · M: Merge

Tip: Applying turns the modifier result into real geometry. Merging removes doubled vertices where branches meet.

Step 7: Solidify & Bevel

Modifiers



Steps:

- Add a Solidify modifier to give the snowflake thickness
- Set the thickness to a small value (e.g. 0.02-0.05)
- Add a Bevel modifier to round sharp edges
- Adjust Bevel segments for smoothness

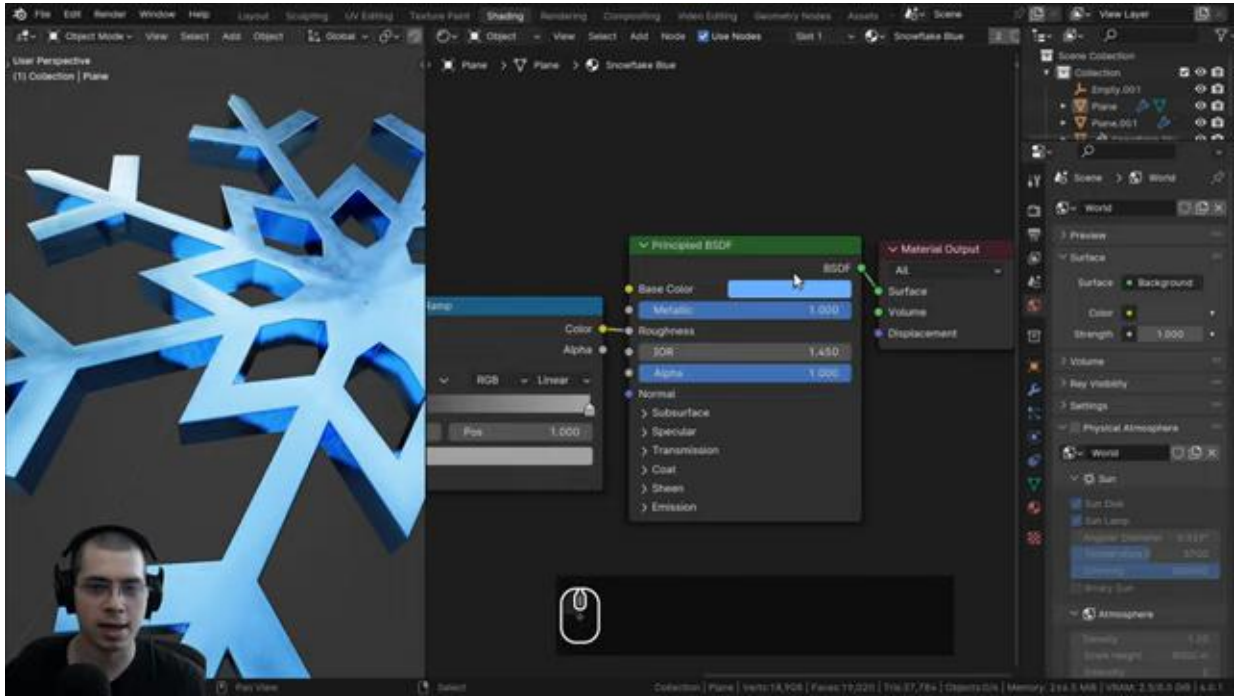
Shortcuts:

Mod properties panel: Add Modifier · Shift + A: still for adding objects

Tip: A flat snowflake looks like paper. Solidify + Bevel make it look like a real 3D object.

Step 8: Material & Color (Optional)

Shading



Steps:

- Switch to the Shading workspace
- Add a new material to the snowflake
- Set Base Color to light blue, white, or gold
- Adjust Metallic and Roughness for an icy or metallic look

Shortcuts:

Z: Shaded preview · F12: Render

Tip: This step is optional. A simple material makes the snowflake look finished.

Creative Challenge: Make It Your Own!

Creative



What to do:

• You followed the tutorial — now use your imagination!

• Design a unique snowflake or winter ornament:

• → Change the number of branches (3, 8, or 12)

• → Add crystals, gems, or holes in the pattern

• → Make a frozen star, snow flower, or icy mandala

• → Combine multiple snowflakes into a scene

• → Try different colors: blue ice, gold metal, glass

• Use modifiers to keep your workflow smart and non-destructive!

• Upload your render to Teams for peer review.

• Most likes wins a 3D-printed prize next class!

Tip: Be original — the best snowflake is one no one has seen before.

Shortcut Cheat Sheet

X	Delete selected	**Shift + A**	Add object (Plane, Sphere,	**Tab**	Toggle Edit / Object Mode
G / S / R	Move / Scale / Rotate	**E**	Extrude	**Ctrl + R**	Loop cut
M	Merge vertices	**Ctrl + J**	Join selected objects	**Ctrl + B**	Bevel selected edges
` (Backtick)	View selector pie menu	**Z**	Toggle Shaded / Wireframe	**F12**	Render image